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DATASCI 224 Final Project

Hidden States, Hidden Disparities: A Hidden Markov Model Analysis of Pain Management Trajectories in LEP Inpatients

Background

Effective pain management depends on clear communication between patients and providers. Limited English Proficiency (LEP), defined as preferring a non-English language and requiring interpreter services, may impair this communication and contribute to disparities in pain assessment and opioid treatment. Prior work has showed that LEP status, cognitive impairment, and hearing impairment are associated with inequities in pain management among older adult inpatients[1]. However, most previous analyses treat each pain assessment as an independent snapshot and miss the within-encounter trajectory. The question remains: whether LEP patients are transiently undertreated or persistently classified in an undertreated state across repeated assessments. This analysis applies Hidden Markov Models (HMM) to characterize these trajectory-level disparities in older hospitalized adults at UCSF.

Data and Study Population

Data were extracted from the UCSF EHR for inpatients aged 65 and older with at least one documented pain assessment between 2020 and 2024. The unit of analysis is the individual pain assessment, with multiple assessments nested within encounters. The analytic sample is restricted to assessments with a pain score greater than zero since opioid decisions are clinically relevant only in the presence of reported pain, yielding 187,397 assessments across 12,137 encounters. LEP patients account for 21.2% of encounters.

Pain severity was measured using the Numeric Rating Scale (NRS, 0–10), the primary self-report instrument at UCSF. Opioid exposure was quantified in morphine milligram equivalents (MME), summed over the 4-hour window following each assessment. Several administration routes were excluded where documented doses do not reliably reflect delivered amounts. These two variables, NRS pain score and 4-hour windowed MME, form the bivariate observation sequence used as input to the HMM at each assessment. Encounter-level variables stratified by LEP status are presented in Table 1. LEP patients in this sample were older on average (79.5 vs 75.5 years), more likely to be female (57.9% vs 49.2%), and predominantly of Asian race (64.8% vs 14.4% among English-speaking patients). LEP patients also received substantially fewer opioids overall, with lower rates of ever receiving an opioid during hospitalization (39.5% vs 51.5%), lower mean total MME per encounter (36.6 vs 212.4), and fewer pain assessments per encounter (7.7 vs 17.5).

Methods

This analysis applies a Gaussian Hidden Markov Model to characterize pain management trajectories across repeated assessments within patient encounters. Two features of the data motivate this choice. First, assessments are naturally sequential as the clinical state at any given assessment plausibly depends on the preceding one. Second, the true clinical state is not directly observed in any assessment t; only the pain score and the opioid dose are recorded. The HMM posits latent states that generate these observations, with the number of states and their clinical meaning determined from the data and labeled post-hoc by inspecting learned emission distributions. HMMs have been previously applied to model inpatient clinical trajectories from EHR data, demonstrating their ability to recover clinically interpretable latent states from routinely collected time-series measurements.[2]

Hidden Markov Model

A Hidden Markov Model is a probabilistic sequence model in which an observed sequence is generated by a hidden sequence of latent states, each depending only on the previous one.[3] Here the observed sequence $X = (x_1, x_2, \ldots, x_T)$ is the series of (pain, MME) measurements at each assessment t, and the latent state $S_t \in \{1, \ldots, K\}$ represents the patient's true clinical condition at that time point. Three parameter sets define the model. The starting distribution $\pi(s) = P(S_1 = s)$ gives the probability of beginning an encounter in state s. The transition matrix $A(i \to j) = P(S_t = j \mid S_{t-1} = i)$ governs how the clinical state evolves between consecutive assessments, with high diagonal values indicating state persistence. The emission distribution $B_s(x_t)$ describes the observation $x_t = (\text{pain}_t, \text{MME}_t)$ generated by each state, modeled as a product of two independent Gaussians under the diagonal covariance assumption[3]:

$$B_s(x_t) = \mathcal{N}(\text{pain}_t;\, \mu^s_{\text{pain}},\, \sigma^{2s}_{\text{pain}}) \times \mathcal{N}(\text{MME}_t;\, \mu^s_{\text{MME}},\, \sigma^{2s}_{\text{MME}}) \tag{1}$$

The Markov assumption $P(S_t \mid S_{t-1}, \ldots, S_1) = P(S_t \mid S_{t-1})$ and output independence assumption $P(x_t \mid S_t, \ldots) = P(x_t \mid S_t)$ together allow the joint probability to factorize as follows, where $\theta = \{\pi, A, B\}$ denotes the full set of model parameters:

$$P(X, S \mid \theta) = \pi(s_1) \times \prod_{t=2}^{T} A(s_{t-1} \to s_t) \times \prod_{t=1}^{T} B_{s_t}(x_t) \tag{2}$$

A six-state model ($K = 6$) was selected for this analysis by minimizing the Bayesian Information Criterion across candidate values of K.

Direct maximization of the observed log-likelihood $\log P(X \mid \theta) = \log \sum_S P(X, S \mid \theta)$ is intractable since summing over all K^T state sequences inside the log has no closed form. Baum-Welch uses Expectation-Maximization, treating hidden states as missing data. The E-step runs a forward pass computing $\alpha_t(s) = P(x_1, \ldots, x_t, S_t = s)$ and a backward pass computing $\beta_t(s) = P(x_{t+1}, \ldots, x_T \mid S_t = s)$, combining them to form soft state assignments

$$\gamma_t(s) = \frac{\alpha_t(s)\,\beta_t(s)}{\sum_{s'} \alpha_t(s')\,\beta_t(s')} \tag{3}$$

and expected transition counts between each pair of states at each time step. The M-step updates parameters in closed form:

$$\pi(s) = \gamma_1(s), \qquad A(i \to j) = \frac{\sum_t \xi_t(i,j)}{\sum_t \gamma_t(i)}, \qquad \mu^s = \frac{\sum_t \gamma_t(s)\,x_t}{\sum_t \gamma_t(s)} \tag{4}$$

where $x_t = (\text{pain}_t, \text{MME}_t) \in \mathbb{R}^2$, so the mean update applies component-wise and the diagonal variance for each component is updated correspondingly.

Baum-Welch is guaranteed to increase log-likelihood at every iteration, converging to a local maximum.[4] As the algorithm may converge to different local maxima depending on initialization, results should be interpreted with this limitation in mind.

Given estimated parameters, Viterbi recovers the most likely state sequence $S^* = \arg\max_S P(X, S \mid \theta)$ for each encounter. Brute force enumeration of K^T paths is intractable; Viterbi achieves exact decoding in $O(K^2T)$ time using dynamic programming. At each step it maintains $\delta_t(s)$, the probability of the most likely path ending in state s at time t[3]:

$$\delta_1(s) = \pi(s) \cdot B_s(x_1) \tag{5}$$

$$\delta_t(s) = \max_{s'} \left[\delta_{t-1}(s') \cdot A(s' \to s)\right] B_s(x_t) \tag{6}$$

A backpointer $\psi_t(s) = \arg\max_{s'} \left[\delta_{t-1}(s') \cdot A(s' \to s)\right]$ records the winning predecessor at each node. We can discard losing paths at every step because the Markov property guarantees two paths arriving at the same state face an identical future, so the weaker path can never recover. The full sequence is recovered by following ψ backwards from $\arg\max_s \delta_T(s)$. All computations are performed in log space to prevent numerical underflow. The decoded sequences assign each assessment a state label, enabling trajectory-level comparison of undertreatment rates between LEP and English-proficient patients.

Results and Discussion

Table 2 presents the learned emission means for each of the six states, along with their post-hoc clinical labels. The states capture a clinically interpretable range of pain management patterns, from State 1 (Undertreated), characterized by moderate pain and near-zero MME, to State 3 (Complex pain management), which reflects the highest pain severity and by far the largest opioid exposure at 199.1 MME on average. The remaining four states represent gradations of treated patients distinguished primarily by opioid dose rather than pain level, as average pain scores across States 0, 2, 4, and 5 are all clustered between 6.7 and 7.3.

Figure 1 presents the proportion of assessments spent in each latent state by LEP status. The largest difference was observed in the undertreated state, where LEP patients spent 63.4% of assessments compared with 53.8% among English-proficient patients, a gap of 9.6 percentage points. This finding suggests that disparities in pain management are not limited to individual assessments but extend to the overall trajectory of care, with LEP patients spending a greater proportion of their encounters in a state characterized by moderate pain and little or no opioid administration.

The disparity ran in the opposite direction for states associated with active opioid treatment. English-proficient patients spent substantially more time in the complex pain management state (2.0% vs 0.2%) and the high-dose treated state (5.7% vs 1.6%), while the standard-dose treated state also showed a modest difference (8.8% vs 5.2%). In contrast, time spent in the low-dose treated state (21.6% vs 23.0%) and moderate-dose treated state (8.0% vs 6.7%) was broadly similar across groups.

Taken together, these results suggest that LEP patients are more likely to remain in a latent state characterized by untreated or minimally treated pain and less likely to transition into states associated with higher levels of opioid treatment. The marked under-representation of LEP patients in the complex and high-dose treatment states further suggests that disparities may become more pronounced at higher levels of opioid titration.

Several limitations warrant acknowledgment. State labels are assigned post-hoc based on emission means and represent clinical interpretations rather than ground truth. The diagonal covariance assumption treats pain and MME as conditionally independent within each state, which may not fully capture their joint distribution. In addition, MME is nonnegative and highly right-skewed, so the Gaussian emission assumption may not fully capture its empirical distribution. The standard HMM also assumes equally spaced observations, which may not hold across all encounters given irregular assessment intervals. Finally, the comparison of state-time proportions between LEP and English-proficient patients is descriptive; formal inference on the group difference would require bootstrap confidence intervals or a permutation test to quantify uncertainty around the 9.6% gap.

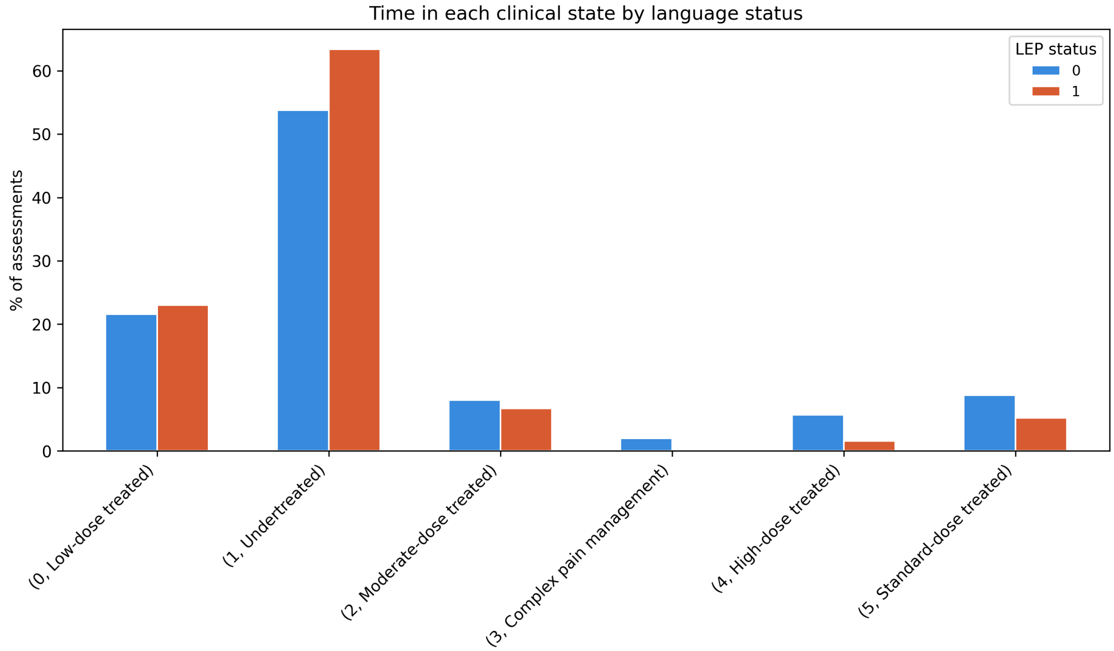
Table 1: Baseline Characteristics at the Encounter Level by LEP Status

Variable	**Overall**	**English-Speaking**	**LEP**
N	12,137	9,570	2,567
Demographics			
Age, mean (SD)	76.3 (8.4)	75.5 (8.0)	79.5 (9.0)
Sex, n (%)			
Male	5,945 (49.0%)	4,865 (50.8%)	1,080 (42.1%)
Female	6,192 (51.0%)	4,705 (49.2%)	1,487 (57.9%)
Race/Ethnicity, n (%)			
White	5,783 (47.6%)	5,485 (57.3%)	298 (11.6%)
Asian	3,042 (25.1%)	1,379 (14.4%)	1,663 (64.8%)
Black or African American	1,272 (10.5%)	1,248 (13.0%)	24 (0.9%)
Latinx	1,148 (9.5%)	708 (7.4%)	440 (17.1%)
Other/Multiracial	892 (7.3%)	750 (7.8%)	142 (5.5%)
Geriatric-related Variables			
Hearing Impairment, n (%)			
No	6,152 (50.7%)	4,950 (51.7%)	1,202 (46.8%)
Yes	5,436 (44.8%)	4,197 (43.9%)	1,239 (48.3%)
Unknown	549 (4.5%)	423 (4.4%)	126 (4.9%)
Dementia/MCI, n (%)	2,261 (18.6%)	1,709 (17.9%)	552 (21.5%)
Pain and Opioid			
Average Pain Score, mean (SD)	5.2 (1.9)	5.2 (1.9)	5.1 (1.9)
Average Assessments, mean (SD)	15.4 (36.9)	17.5 (40.5)	7.7 (16.0)
Total MME, mean (SD)	175.2 (1607.5)	212.4 (1806.4)	36.6 (165.7)
Ever Received Opioid, n (%)	5,940 (48.9%)	4,926 (51.5%)	1,014 (39.5%)
Prior Opioid Use, n (%)	4,671 (38.5%)	3,948 (41.3%)	723 (28.2%)

Table 2: Learned emission means for each HMM state on the original scale, with post-hoc clinical labels assigned based on average pain score and MME.

State	**Clinical Label**	**Avg Pain**	**Avg MME**
0	Low-dose treated	6.7	7.2
1	Undertreated	5.2	0.0
2	Moderate-dose treated	6.8	15.0
3	Complex pain management	7.7	199.1
4	High-dose treated	7.3	48.5
5	Standard-dose treated	7.2	22.2

Figure 1: Time spent in each clinical state by language status. Blue bars represent English-proficient patients (LEP=0); orange bars represent LEP patients (LEP=1).



References

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